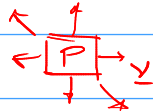
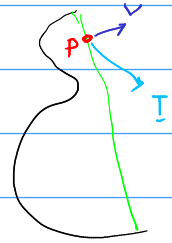
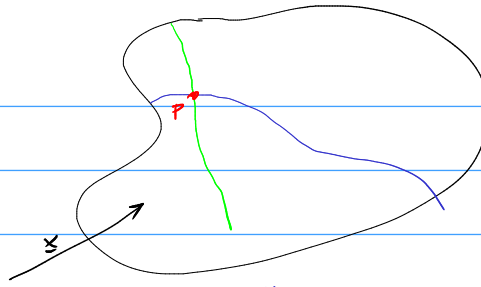


$$\underline{\underline{\Gamma(x)}}$$

$\underline{\underline{T}}$



Fórmula Cauchy:

$$\underline{\underline{T}} = \underline{\underline{\Gamma}} \cdot \underline{\underline{v}}$$

$$\underline{\underline{T}} = \underline{\underline{v}} \cdot \underline{\underline{\Gamma}} \rightarrow T_i = v_j \Gamma_{ji}$$

$$T_i = \Gamma_{ij} v_j$$

$$\|\underline{\underline{v}}\| = 1$$

$$\Gamma_{ij} = \Gamma_{ji}$$

$$\begin{bmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{21} & \Gamma_{22} \end{bmatrix}$$

$$\underline{\underline{T}} = \begin{bmatrix} \Gamma_{12} \\ \Gamma_{22} \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\underline{\underline{T}} = \begin{bmatrix} \Gamma_{11} \\ \Gamma_{21} \end{bmatrix}$$

$$\underline{\underline{x}} \cdot \underline{\underline{y}} = \alpha$$

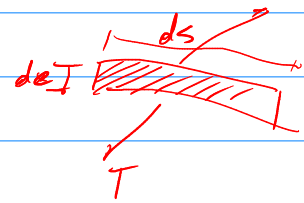
$$\underline{\underline{A}} \cdot \underline{\underline{x}} = \underline{\underline{y}}$$

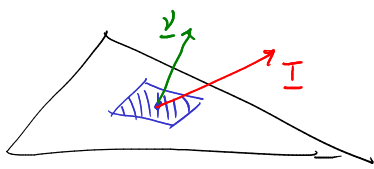
$$\underline{\underline{C}} : \underline{\underline{\epsilon}} = \underline{\underline{\sigma}}$$

$$\bar{A}_{11} x_1 + \bar{A}_{12} x_2 + \bar{A}_{13} x_3 = \gamma_1$$

$$\bar{A}_{21} x_1 + \dots = \gamma_2$$

$$\vdots$$





$$\underline{T} = \underline{y} \cdot \underline{y}$$

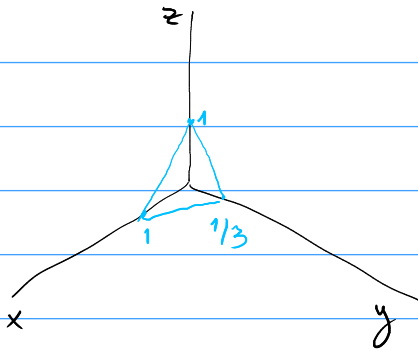
$$\begin{matrix} x & y & z \\ x & \begin{pmatrix} 0 & 1 & 2 \end{pmatrix} \\ y & \begin{pmatrix} 1 & 2 & 0 \end{pmatrix} \\ z & \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} \end{matrix}$$

$$x+3y+z=1$$

$$\underline{p} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \underline{y}$$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

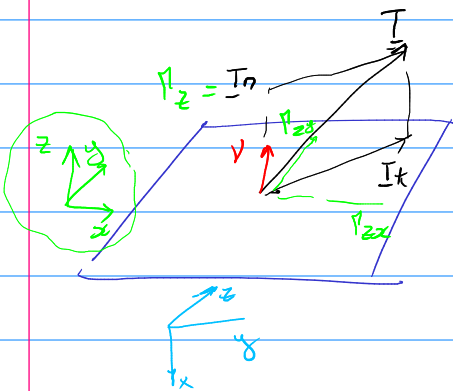
$$1(x-1) + 3(y-0) + 1(z-0) = 0$$



$$\underline{y}^P = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{y} = \frac{\underline{y}^P}{\|\underline{y}^P\|} = \frac{\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}}{\sqrt{1^2 + 3^2 + 1^2}} = \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{T} = \underline{y} \cdot \underline{y} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix}$$



Projection

$$T_n = \underline{T} \cdot \underline{y} = \frac{1}{\sqrt{11}} \begin{bmatrix} 5 \\ 7 \\ 3 \end{bmatrix} \cdot \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{29}{11}$$

Vector normal

$$\underline{T}_n = T_n \underline{y} = \frac{29}{11\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{T} = \underline{T}_n + \underline{T}_t \Rightarrow \underline{T}_t = \underline{T} - \underline{T}_n = \frac{1}{11\sqrt{11}} \begin{bmatrix} 26 \\ -10 \\ 4 \end{bmatrix}$$

